The Making and Breaking of Radioactive Molecules

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Radioactive Molecules at NEPTUNE

- The Standard Model fails to describe 95% of the mass and energy in the Universe, matter-antimatter imbalance, …
- Our group is searching for violations of fundamental symmetries in radioactive molecules [1]. We work in close collaboration with Prof. Garcia (MIT), Prof. DeMille (JHU), Prof. Blaum (MPIK), Prof. Ringle (FRIB), Prof. Fan (Harvard), Prof. Hutzler (Caltech), Prof. Dilling (JLAB) on:

NEPTUNE: Trapping (radioactive) molecular ions in a Penning ion trap can amplify electroweak nuclear effects (e.g., nuclear anapole moment) by more than **12 order's [2]!**

$$W_{PV} \sim \frac{<-|H_{PV}|+>}{\boxed{E_- - E_+}}$$

Atom:	~1 eV
Molecule:	~10 µeV
Molecule + ***B***:	~1 peV

Factor >10¹² amplification!

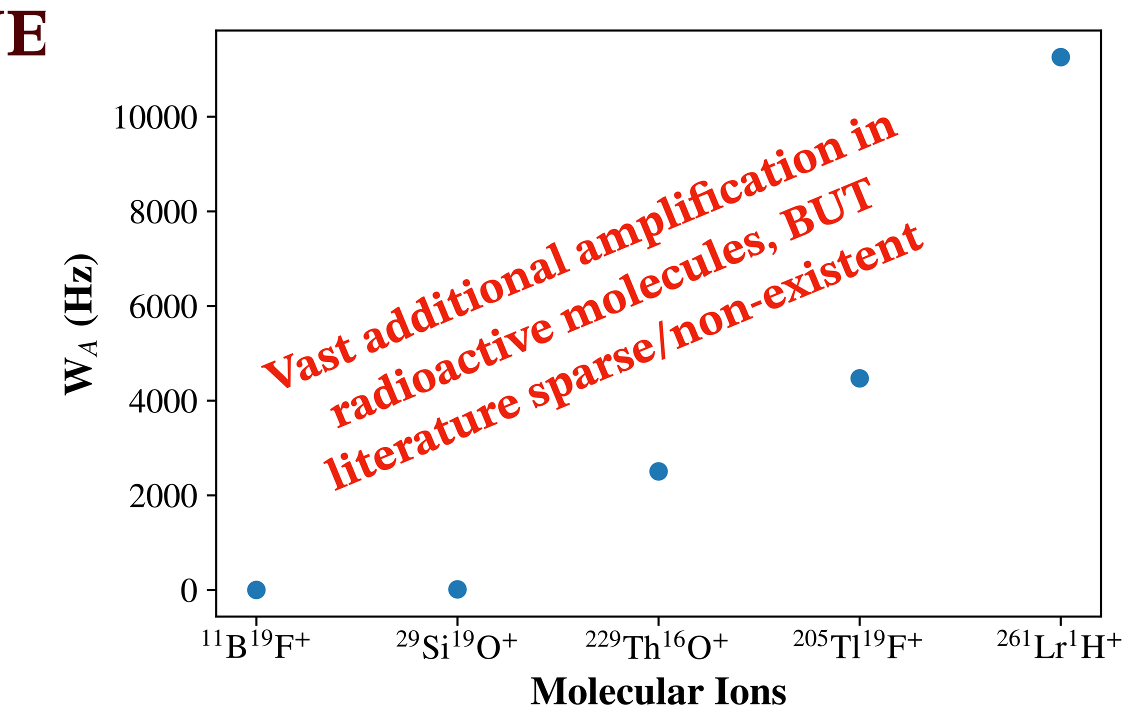

Fig 1. Diatomic molecular ions with sizable weak matrix elements W_A [1]

Piezo-based Laser Ablation Ion Source

- **Stable and long-lived molecules** in vacuum can be created using laser ablation of solid targets
- Ion acceleration with noble gases and rapid expansion into vacuum creates **"cold" molecules** in the lowest ro-vibrational states
- Current designs often use solenoid gas valves, leading to long (>100 µs) gas pulses and, thus, high gas loads that reduce cooling efficiency
- Piezoelectric valves [3] allow for **ultra-short gas pulses** (~10 µs)
- Design and commissioning of the ion source in collaboration with Prof. Yan Zhou (UNLV)

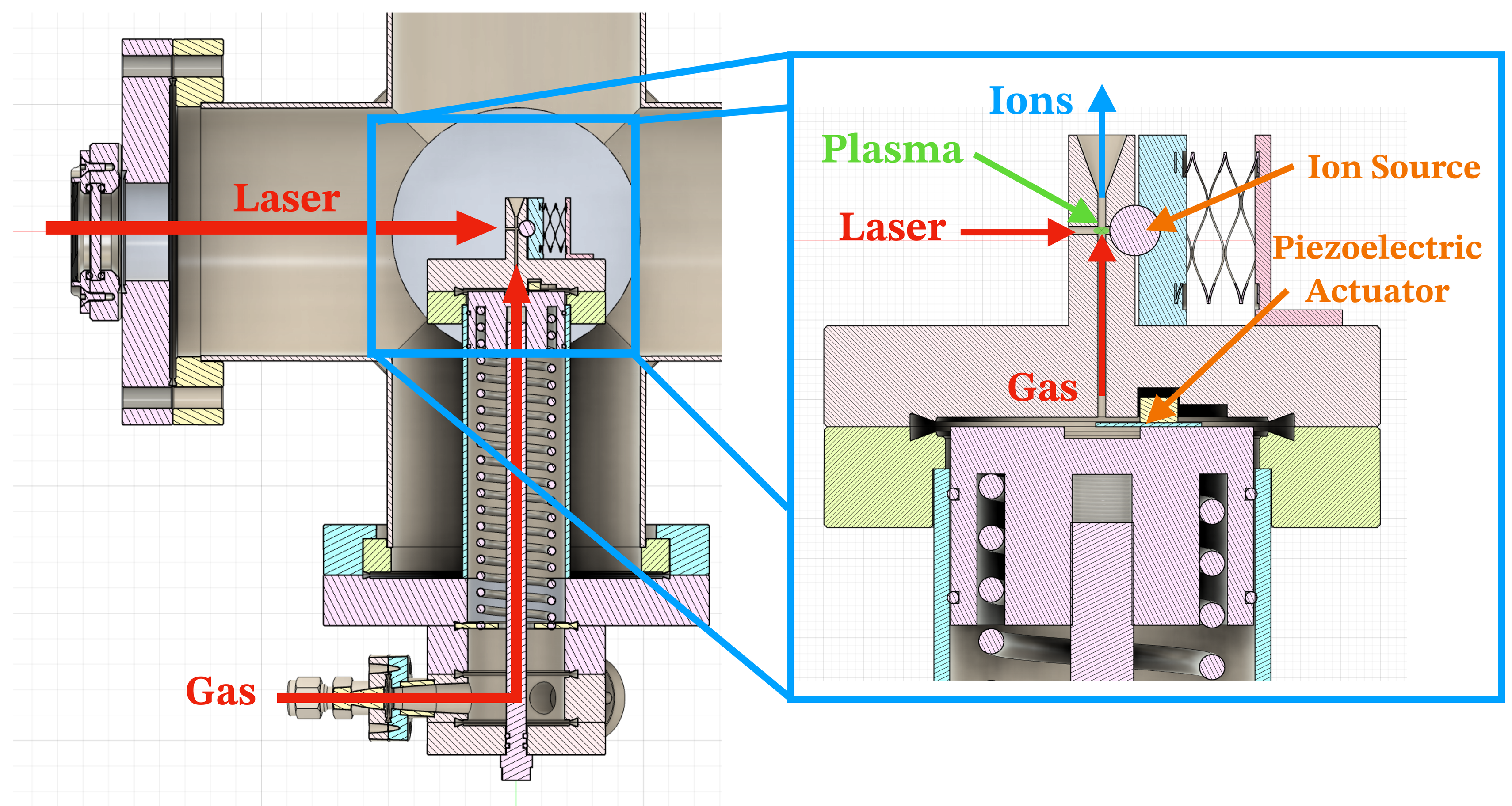

Fig 2. A three-dimensional cross section of the Piezoelectric Laser ablation ion source

The Si_3N_4 Dissociation Window

- **2 mm × 2 mm** Si_3N_4 window, just **~21 nm thick** (≈ 70 atomic layers)
- Impinging molecular beam get ro-vibrationally excited through collisions far above the molecular **dissociation** threshold
- Withstands pressure differentials up to 200 mbar → allows for gas-filled **RFQ trapping section** after dissociation for efficient cooling of ***isotopes of interest*** *OR* the creation of ***molecules of interest***

Fig 5. The Si_3N_4 window installed on a CF flange [6].

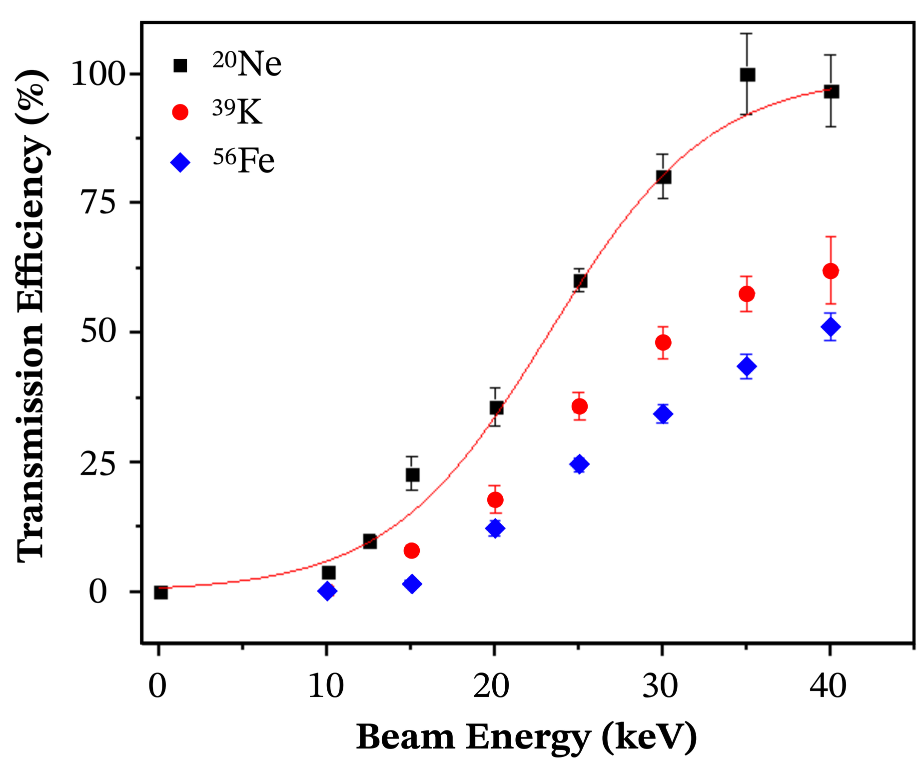

Fig 6. First transmission efficiency results of the CID-GC [6].

Collision-Induced Dissociation Gas Cell (CID-GC)

- Short-lived molecular ions require radioactive beam facilities
- Beam facilities do not typically provide the molecules of interest, while molecular contaminants and isotopes of interest trapped in molecules pose an increasing challenge to stopped beam experiments

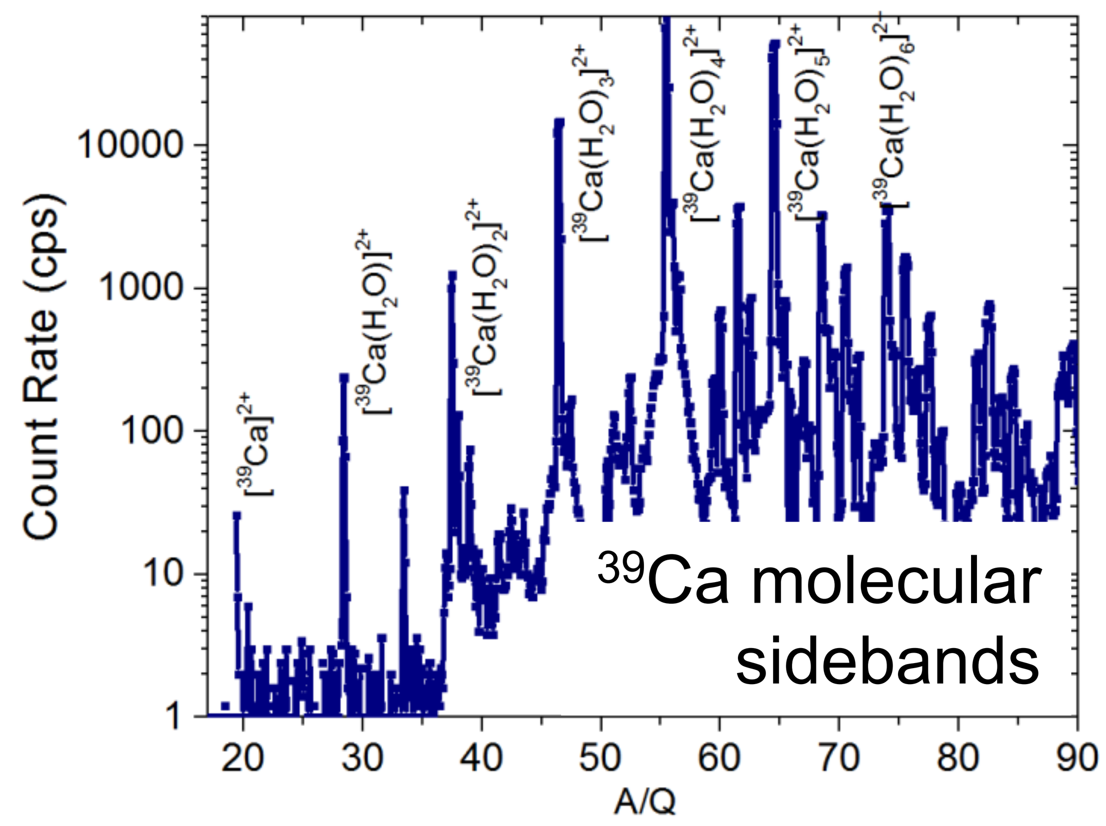

Fig 3. Example of a molecular contaminant spectrum (Courtesy to R. Ringle).

- Various groups (e.g. at GSI [4] and TRIUMF [5]) have applied CID in RFQ cooler/bunchers with varying degrees of success and efficiency, while laser dissociation is molecule specific and requires extensive knowledge of the molecule

→ With Prof. Ringle at FRIB, we are developing a general-purpose device to **break up molecules** through collision-induced dissociation [6]

→ Molecules pass through an ultra-thin Si_3N_4 membrane to induce dissociation, followed by RFQ for trapping and/or molecule formation

→ Molecular spectroscopy performed at Texas A&M in new general-purpose action spectrometer ***(see poster of Nick Schnoor)***

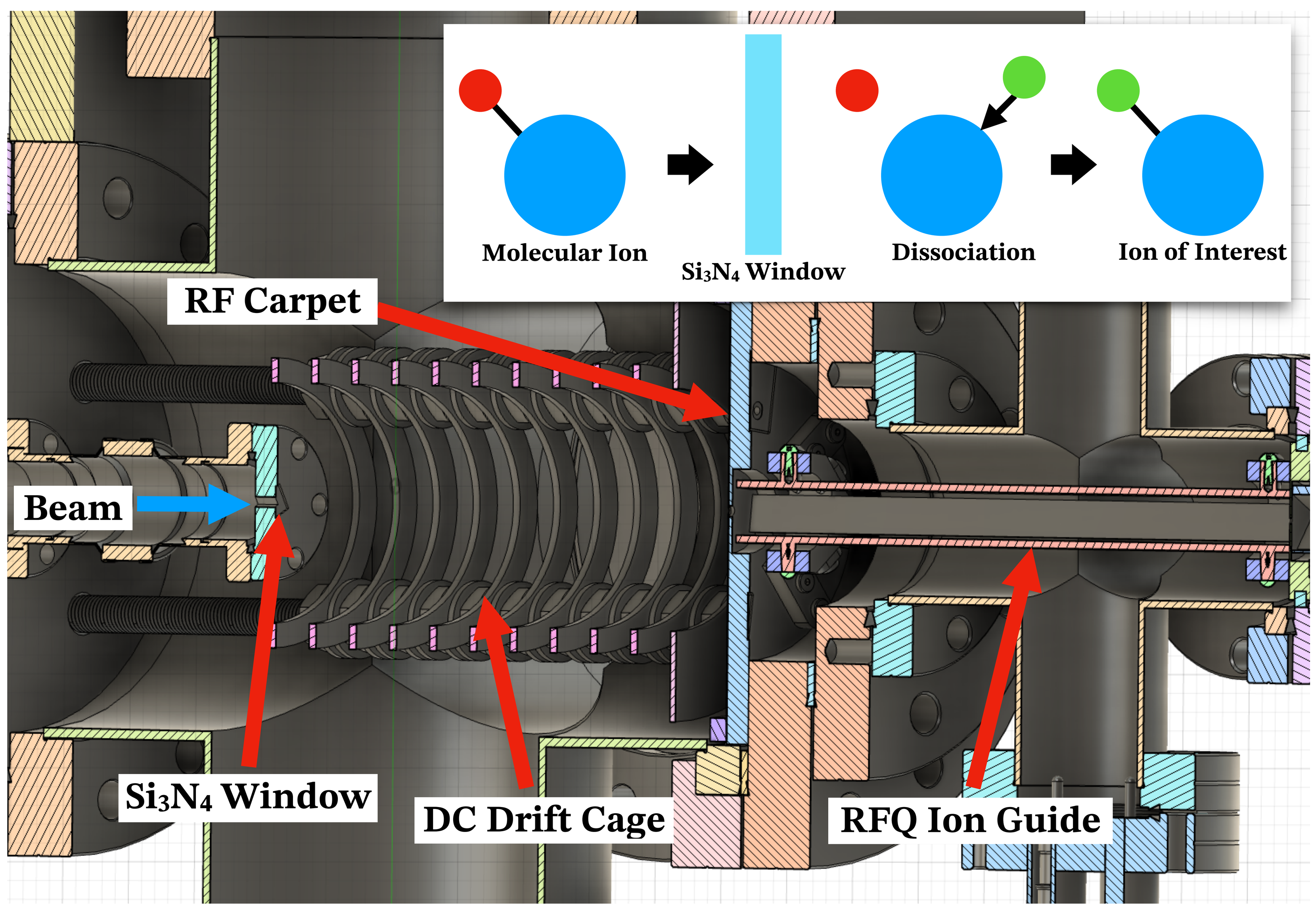

Fig 4. Cross-section of the CID-GC [6] and illustration of the dissociation + formation mechanism (inset).

References

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[5] Jacobs A et al. *Int. J. Mass Spec.*, **482**, 116931 (2022).
[6] Ringle R. Final report: DE-SC0021423 (2024).

Acknowledgment

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